Arc Length and Improper Integrals

ANSWER KEY

Arc length

1 Find the arc length of $y=\frac{2}{3}x^{3/2}$ from $x=0$ to $x=3$.
$y'=x^{1/2}$, so $L=\int_0^3\sqrt{1+x}\,dx=\left[\frac{2}{3}(1+x)^{3/2}\right]_0^3=\frac{2}{3}(8-1)=\frac{14}{3}$.

2 Set up (do not evaluate) the arc length integral for $y=\sin x$ on $[0,\pi]$.
$L=\int_0^{\pi}\sqrt{1+\cos^2 x}\,dx$.

Improper integrals

3 Evaluate $\int_1^{\infty}\frac{1}{x^2}\,dx$, or state that it diverges.
$\lim_{b\to\infty}\left[-\frac{1}{x}\right]_1^b=\lim_{b\to\infty}\left(1-\frac{1}{b}\right)=1$: converges to 1.

4 Evaluate $\int_1^{\infty}\frac{1}{x}\,dx$, or state that it diverges.
$\lim_{b\to\infty}[\ln x]_1^b=\lim_{b\to\infty}\ln b=\infty$: diverges.

5 Evaluate $\int_0^1\frac{1}{\sqrt{x}}\,dx$ (improper at the lower limit).
$\lim_{a\to 0^+}[2\sqrt{x}]_a^1=2-0=2$: converges to 2.

More arc length practice

6 Find the arc length of the line $y=3x+1$ from $x=0$ to $x=4$, and verify your answer using the distance formula.
$y'=3$: $L=\int_0^4\sqrt{1+9}\,dx=4\sqrt{10}$. Distance check: endpoints $(0,1)$ and $(4,13)$:
$\sqrt{4^2+12^2}=\sqrt{16+144}=\sqrt{160}=4\sqrt{10}$. ✓

7 Set up (do not evaluate) the arc length integral for $y=\ln x$ on $[1,e]$.
$y'=\frac{1}{x}$: $L=\int_1^e\sqrt{1+\frac{1}{x^2}}\,dx$.

More improper integrals

8 Evaluate $\int_1^{\infty}\frac{1}{x^{1.5}}\,dx$, or state that it diverges.
$\lim_{b\to\infty}[-2x^{-0.5}]_1^b=\lim_{b\to\infty}\left(2-\frac{2}{\sqrt{b}}\right)=2$: converges to 2.

9 Evaluate $\int_{-\infty}^{0}e^x\,dx$, or state that it diverges.
$\lim_{a\to-\infty}[e^x]_a^0=1-\lim_{a\to-\infty}e^a=1-0=1$: converges to 1.

The p-integral test

10 For $\int_1^\infty \frac{1}{x^p}\,dx$, state the condition on p for convergence, and explain why using the antiderivative $\frac{x^{1-p}}{1-p}$ (for $p \neq 1$).

Converges when $p > 1$. As $b \to \infty$, $b^{1-p} \to 0$ only when $1 - p < 0$, i.e. $p > 1$; otherwise the limit is infinite (or, for $p = 1$, the antiderivative is $\ln x$, which also diverges).

11 Use the p-integral result to quickly determine whether $\int_1^\infty \frac{1}{x^{0.9}}\,dx$ converges.

$p = 0.9 < 1$: diverges.

Improper integrals with infinite discontinuities inside the interval

12 Evaluate $\int_0^2 \frac{1}{(x-1)^2}\,dx$, or explain why it diverges (note the integrand is undefined at $x = 1$, inside the interval).

Split at $x = 1$: $\int_0^1 \frac{1}{(x-1)^2}dx + \int_1^2 \frac{1}{(x-1)^2}dx$. Near $x = 1$ from either side, $\lim_{b \to 1}\left[-\frac{1}{x-1}\right]$ blows up to $\pm\infty$: both pieces diverge, so the full integral diverges.

13 Evaluate $\int_{-1}^1 \frac{1}{x^{1/3}}\,dx$ (an odd function with a discontinuity at $x = 0$), splitting at $x = 0$.

By symmetry (odd integrand), $\int_{-1}^0 x^{-1/3}dx = -\int_0^1 x^{-1/3}dx$ (each piece converges since $p = \frac{1}{3} < 1$ for this type of integrand near a finite discontinuity). The two pieces cancel: the integral equals 0.